#1154 (1571324785): *Lightweight Q-Learning Based Joint RACH Optimization for Ultra-Dense NB-IoT Smart Meter Networks*

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Authors	Hide authors Sandeep Kumar (Indian Institute of Information Technology, Kota, India)
Full paper title	*Lightweight Q-Learning Based Joint RACH Optimization for Ultra-Dense NB-IoT Smart Meter Networks* Only the chairs can edit
Conference and track	**2026 5th Edition of IEEE Delhi Section Flagship Conference (DELCON)** - *Track 1: Smart Communication and Networking Technologies*
Abstract	Only the chairs can edit Demand-response events in NB-IoT-enabled Advanced Metering Infrastructure (AMI) can trigger...
Keywords	Narrowband Internet of Things (NB-IoT); Massive Machine-Type Communications (mMTC); Random Access Channel (RACH); Q-Learning; RACH Congestion Control Only the chairs can edit
Topics	*Smart Communication and Networking Technologies*: Resource Allocation and QoS; Network Performance Optimization; Multiple Access and AI in Next Generation Wireless Technology. (Only the chairs can edit)
Personal notes	
Roles	You are the creator and an author for this paper.
Status	Rejected

Review Manuscript PDF	Final manuscript

Double Blind Review

Rev.	Relevance and timeliness	Novelty and originality	Technical content and scientific rigour	Quality of presentation	Double-blind?
A	Limited 2	Minor variations on a well-investigated subject. 2	Marginal work and simple contribution. Some flaws. 2	Substantial revision work is needed. 2	Yes (This paper follows all the double-blind submission guidelines)

Strong aspects

Timely work and practically relevant problem statement, but limited contribution as compared to the state of the art.

Weak aspects

The experimental comparison is internally inconsistent. Section V-A lists five baselines (3GPP Static, LE-URC, Smart ACB-only, Single-Parameter RL, Dual-Parameter RL); Table II reports a different set (adds High-ACB Fixed and Heuristic ACB+BI, drops both RL baselines); Figs. 3-4 legends show a third set (DQN, DDQN, PPO). These three sets don't reconcile.
Fig. 4's caption calls the deep-RL curves "DRL proxies," which suggests DQN/DDQN/PPO were approximated rather than implemented, if so, the claim of outperforming DRL is not established on equal footing.
In Table II, Pc and Ps are exact complements in every row (Pc = 1 − Ps), so the collision column carries no independent information, and the metric "e" is never defined.
Presentation defects: "Error! Reference source not found." at the architecture figure,
"[Rxx]" placeholder for LE-URC (should be [13]),
"Fig. 2illustrates" spacing.
Novelty is incremental over prior tabular Q-learning for RACH.

Recommended changes

PI fix below issues:
The experimental comparison is internally inconsistent. Section V-A lists five baselines (3GPP Static, LE-URC, Smart ACB-only, Single-Parameter RL, Dual-Parameter RL); Table II reports a different set (adds High-ACB Fixed and Heuristic ACB+BI, drops both RL baselines); Figs. 3-4 legends show a third set (DQN, DDQN, PPO). These three sets don't reconcile.
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B	Acceptable 3	Minor variations on a well-investigated subject. 2	Marginal work and simple contribution. Some flaws. 2	Unacceptable. 1	Yes (This paper follows all the double-blind submission guidelines)

Strong aspects

Please refer to the 'Recommended changes' section for a detailed explanation. The same argument is not repeated here.

Weak aspects

Rev.	Relevance and timeliness	Novelty and originality	Technical content and scientific rigour	Quality of presentation	Double-blind?

Please refer to the 'Recommended changes' section for a detailed explanation. The same argument is not repeated here.

Recommended changes

The paper suffers heavily from unplanned AI usage and poor presentation. Although the motivation is relevant, the actual message of the paper and the novelty is blurred due to such a messy presentation.

(1) Let's start with Fig. 2, which is supposed to summarize the entire workflow. The figure is clealry AI generated and this is why there is no connection between step 5 and step 6. In fact, this flaw is imbibed inside the description of the workflow. We can't find any description of the 'compute-reward' step in the text.

(2) 'The complete learning procedure is summarised in Algorithm 1' .. where is it?

(3) 'Error! Reference source not found.' .. please read your own paper before submission.

(4) '$\mathbf{1}(\cdot)$ is the indicator function' .. another classic example of AI hallucination.

(5) Where is the rest of Fig. 4? It is trimmed on the right side.

There are numerous such flaws throughout. Unless they are taken care of, this paper remains unreadable.

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